

Reproduction of DeReFusion and Diagnostics of Operator Specialization for Financial Time Series*

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Abstract

This report consolidates the work completed up to 2026-09-12 on the reproduction of DeReFusion and on a sequence of controlled diagnostics that ask a single question: does the structural state of the input determine which operator is more useful? We first reproduce the core comparison of the original paper (DLinear base branch, LSTM–Transformer residual branch, parameter-free additive fusion, all wrapped in RevIN) on the S&P 500 and two further assets. We then stratify forecast errors by causal relative volatility and find that the marginal value of the nonlinear branch is *asset-dependent* rather than universal: it is significantly positive for the S&P 500 (and seed-robust), directionally positive but statistically inconclusive for ETHUSD, and significantly *reversed* for BTCUSD. A capacity-sensitivity experiment (hidden width $23 \rightarrow 128$, everything else frozen) shows that the earlier “linear dominates everywhere” reading was partly an artefact of a narrow nonlinear bottleneck: with adequate capacity ETHUSD prefers the nonlinear operator in 36/36 observations, while BTCUSD never does. A capacity-matched proxy routing experiment (12 pre-registered structural states, linear vs. MLP, identical protocol) finds no robust state-level operator specialization. Finally, an exploratory sweep over nine assets shows interaction effects that span zero (six negative, three positive; five statistically supported) and two leave-one-out-stable associations (realized volatility and $|\text{ACF}_1|$), which we report as a *candidate* asset-level structural regularity only. We state explicitly what the evidence excludes (sample-level routing), what it supports (asset-level

*Reproduction of Hsieh and Chen (2026), Applied Soft Computing 203, 116252, extended with controlled diagnostics of regime- and asset-dependent operator suitability.

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operator heterogeneity), and what remains unresolved.

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1. Scope and research questions

The work reported here follows an evidence-first protocol: reproduce the reference result, then ask one question at a time, with every grouping rule, threshold and operator definition fixed *before* looking at outcomes. Three questions structure the report.

- **Q-A (reproduction).** Does the published ordering of fusion strategies (additive fusion > linear base > learnable gate) reproduce on independently fetched data?
- **Q-B (regime).** Is the marginal contribution of the nonlinear branch conditioned on the volatility regime of the input window?
- **Q-C (specialization).** Does the *structural state* of the input determine which operator is more useful — i.e. is there an experimental basis for routing?

All experiments run on CPU, use a frozen 70/10/20 chronological split, and report paired bootstrap confidence intervals (4000 resamples) with paired win rates. No hyper-parameter was tuned after observing test outcomes.

2. Reproduction of the reference comparison

The dataset of the original repository is not redistributable, so ten daily OHLC series (2016–2025) were re-fetched from Yahoo Finance; row counts match the paper (2513 for indices and equities, 2602 for FX, 3653 for BTCUSD, 2975 for ETHUSD). Table 1 reports the S&P 500 comparison at horizons $T \in \{1, 24\}$ (seed 2021).

At $T = 24$ the published ordering reproduces: additive fusion outperforms the linear base branch by 11.1% in MSE, while both gate variants are worse than parameter-free addition. At $T = 1$ the linear branch is marginally better, in line with the “decomposition-then-simple-model” behaviour of long-horizon linear forecasters. Extending the same protocol to two further assets at

Table 1: Reproduction of the core comparison (GSPC, seed 2021, normalized scale).

Model	$T=1$ MSE	$T=24$ MSE	$T=24$ R^2	Params
DeReFusion (additive)	0.01647	0.06230	0.8691	28,436
revin-DLinear (linear)	0.01577	0.07007	0.8528	4,664
gatev1 (volatility-aware)	—	0.06767	0.8578	28,580
gatev2 (learnable)	0.02526	0.07191	0.8489	~28k

Table 2: Stratified performance, relative volatility ($\Delta = \text{MSE}_{nonlin} - \text{MSE}_{lin}$; negative favouring the nonlinear branch). [†]CI excludes zero.

Asset	low 20%	high 50%	high 20%	Interaction [95% CI]
GSPC (2021)	+6.0%	+14.2% [†]	+21.6% [†]	-0.01205 [-0.01545, -0.00876] [†]
GSPC (2022)	-2.7%	+23.2% [†]	+29.6% [†]	-0.02248 [-0.02719, -0.01792] [†]
ETHUSD	+16.7% [†]	+6.3% [†]	+14.2% [†]	-0.00943 [-0.01881, +0.00033]
BTCUSD	+3.5%	-1.0%	-3.5%	+0.01502 [+0.00615, +0.02378] [†]

$T = 24$ gives DeReFusion 0.21797 vs. DLinear 0.22322 for BTCUSD (+2.4%) and 0.14602 vs. 0.14876 for ETHUSD (+1.8%); the additive advantage is therefore ordered $\text{GSPC} \gg \text{BTCUSD} \approx \text{ETHUSD}$.

3. Volatility-regime stratification

Each test window is assigned a regime from a *causal* volatility measure computed on the input window only: the standard deviation of log returns, and a de-trended version, $RV_t^{rel} = RV_t / \text{median}(RV_{t-120:t-1})$. Table 2 reports the relative improvement of the nonlinear branch over the linear base per stratum (relative volatility, the primary measure) plus the interaction effect $\Delta_{high50} - \Delta_{low50}$.

Two findings follow. First, the S&P 500 effect is *seed-robust*: both seeds give a significantly negative interaction of the same order, and the calm-regime strata are indistinguishable between branches. Second, BTCUSD reverses the sign significantly, so “nonlinear advantage grows with volatility” cannot be stated as a universal criterion. A methodological check explains part of the earlier confusion: the raw volatility measure is strongly confounded with time (correlation of RV with the sample index: +0.44 for GSPC, -0.67 for BTCUSD). The de-trended RV^{rel} removes that trend; under it the spurious

Table 3: Capacity sensitivity. Mean ΔMSE (negative = nonlinear better); in parentheses the number of the 36 (state, seed) observations preferring the nonlinear operator.

Hidden	MLP params	GSPC	BTCUSD	ETHUSD
23	9,431	+0.0598 (0/36)	+0.1783 (0/36)	+0.0145 (12/36)
64	26,200	+0.0250 (7/36)	+0.1888 (0/36)	− 0.0281 (36/36)
128	52,376	+0.0061 (8/36)	+0.2998 (0/36)	− 0.0321 (36/36)

“linear wins in the middle quintile” of GSPC disappears. Reported figures therefore use RV^{rel} as the primary measure.

4. Adaptive fusion (gating) does not pay

The natural use of the regime structure would be a gate $\alpha(X)$ mixing the two branches. The framework already provides exactly this variant, and we ran it on the asset where the structure is strongest and seed-robust (GSPC, $T = 24$): the volatility-aware gate obtains MSE 0.06767, i.e. 8.6% *worse* than the parameter-free additive fusion (0.06230) and only 3.4% better than the plain linear base (0.07007). Since the nonlinear branch is never worse than the linear one in the calm strata, the optimal gate weight is essentially constant at one, leaving no switching headroom for the gate to exploit. Together with the previously reported negative results for the learnable gate, this motivates stopping the adaptive-fusion line of work rather than building a router on top of it.

5. Capacity sensitivity: the rigidity was partly an artefact

The proxy experiment of the next section compares a linear operator with a capacity-matched MLP (9,240 vs. 9,431 parameters, +2.1%). That matching forces a 384-dimensional input through a 23-dimensional hidden layer — a severe bottleneck. We therefore re-ran the structural-state evaluation with the hidden width as the *only* variable ($23 \rightarrow 64 \rightarrow 128$; data, features, splits, optimizer, learning rate, batch size, epochs, early stopping, seeds and evaluation protocol unchanged). Table 3 reports, per width and asset, the mean ΔMSE over 12 structural states and 3 seeds.

The three pre-registered checks give a consistent reading. (i) *Sign reversals*: 25 state-level reversals between widths, of which 23 belong to ETHUSD

(all three seeds) and 8 to GSPC (all of them at seed 2023 only, i.e. seed-unstable); BTCUSD has none. (ii) *Spread across states*: the range of ΔMSE across states *shrinks* with capacity for GSPC ($0.106 \rightarrow 0.060$) and ETHUSD ($0.089 \rightarrow 0.040$); the BTCUSD increase is driven by the magnitude of a single high-error state. (iii) *Per-seed direction consistency* falls from 31/36 to 29/36 to 28/36. Taken together: the ETHUSD preference for the nonlinear operator is *asset-wide* once capacity is sufficient — all twelve states reverse, not a subset — so it is asset-level heterogeneity rather than state-level specialization. Accordingly we record the earlier “linear dominates everywhere” statement as partly capacity-driven, and we do not treat the capacity sweep as evidence of routing.

6. Structure-aware routing: a minimal controlled proxy experiment

6.1. Design

Twelve structural states were defined *before* any outcome was inspected, using causal window features (ACF at lags 1/5/10, a trend t -statistic, sign persistence, a jump ratio with a fixed threshold $c = 3\sigma$, the de-trended relative volatility, skewness and kurtosis). Binary states use train-set medians as thresholds; a composite *structure strength* state and a $K=3$ k -means cross-check are also reported. The operators are a linear map (9,240 parameters) and an MLP with one 23-unit hidden layer (9,431 parameters), trained under an identical protocol (same seed, optimizer, steps, batch, learning rate, early stopping, preprocessing).

6.2. Result

For GSPC and BTCUSD the linear operator is better in *every* one of the twelve states at matched capacity (GSPC ΔMSE between +0.047 and +0.081, significant in all three seeds; BTCUSD between +0.129 and +0.448). Only ETHUSD shows between-state variation of both signs (structure-weak states favouring the nonlinear operator by up to -0.013 , structure-strong states favouring the linear one by +0.033). No state shows a preference that survives the capacity sweep. **Main finding: no robust operator specialization is established at capacity-matched scale.** The one cross-asset invariant is that the two operators are indistinguishable in calm regimes.

Table 4: Asset-level interaction effects (relative volatility). [†]CI excludes zero.

Asset	$\Delta_{interaction}$	95% CI	Supported?
EURUSD	-0.1371	$[-0.164, -0.111]$	yes [†]
GSPC	-0.0173	$[-0.027, -0.009]$	yes [†]
USDJPY	-0.0122	$[-0.026, +0.002]$	no
ETHUSD	-0.0094	$[-0.019, +0.000]$	no
DJI	-0.0084	$[-0.019, +0.003]$	no
TM	-0.0081	$[-0.028, +0.011]$	no
BTCUSD	+0.0150	$[+0.006, +0.024]$	yes [†]
BABA	+0.0405	$[+0.022, +0.060]$	yes [†]
NVO	+0.0976	$[+0.046, +0.149]$	yes [†]

Table 5: Exploratory Spearman association of asset structure features with $\Delta_{interaction}$ ($N = 9$) and LOO stability.

Feature	ρ	p	LOO range	Stable?
$ \text{ACF}_1 $	+0.683	0.042	$[+0.548, +0.810]$	yes
Realized volatility	+0.633	0.067	$[+0.476, +0.857]$	yes
Kurtosis	+0.533	0.139	$[+0.333, +0.643]$	yes
Trend persistence	+0.100	0.798	$[-0.286, +0.381]$	no (sign flips)
Jump ratio	—	—	—	constant (0.0105)

7. Asset-level operator heterogeneity across nine assets

Finally, the same frozen protocol ($T = 24$, seed 2021) was run on seven additional assets to map the *sign* of the interaction effect. GSPC aggregates its two seeds into a single asset, so $N = 9$. Table 4 reports the interaction effect with its confidence interval, and Table 5 the exploratory association between asset-level structure features and the interaction effect, including a leave-one-asset-out (LOO) check.

The sign of the effect spans zero: six assets are negative, three positive, with five statistically supported. Two features are LOO-stable: assets with higher realized volatility and stronger first-order autocorrelation show a *less* favourable (or reversed) nonlinear advantage in turbulent regimes. **These associations are exploratory and are reported as a candidate regularity only** — with nine assets, a single p -value is not decisive, and the jump-ratio feature turns out to be constant across all assets at the median

Table 6: Consolidated answers to the five diagnostic questions.

Question	Answer
Q1 Robust sample-level routing evidence?	No. Neither the routing proxy nor the capacity sweep produces a stable state $\rightarrow$ operator mapping.
Q2 Genuine asset-level operator heterogeneity?	Yes. At adequate capacity ETHUSD prefers the nonlinear operator in 36/36 observations, BTCUSD prefers linear at every capacity, GSPC is near parity.
Q3 Explainable by the existing structural features?	Only as a candidate: realized volatility and $ \text{ACF}_1 $ associate with the interaction sign and survive LOO; the sign spans zero across nine assets.
Q4 Principal remaining uncertainty?	Asset dependence and operator form — not the structural representation, and not routing, which is now largely excluded.
Q5 May a stronger nonlinear operator be reconsidered?	Only as one candidate, tested against a capacity-matched MLP, and not by extrapolating “ETHUSD prefers nonlinear” to any specific operator family.

(one exceedance in 95 returns), i.e. it carries no discriminative information at the pre-registered threshold. Trend persistence flips sign under LOO and is marked *single-asset sensitive*.

8. Diagnosis and decisions

The practical decisions recorded here are: (i) stop the adaptive-fusion line — a volatility gate measures worse than parameter-free addition, and the structure it would exploit leaves no switching headroom; (ii) do not build a neural router — the state $\rightarrow$ operator mapping is not established; (iii) treat operator choice as an asset-level configuration question, which is what the evidence supports; and (iv) keep any structured nonlinear operator conditional on a reproducible context and on a capacity-matched comparison.

9. Limitations

Operator fidelity. The proxy operators (a linear map and an MLP over flattened windows) differ from the branches of the framework, which combine a DLinear decomposition with an LSTM–Transformer residual branch under RevIN. Consequently, the proxy results must not be read as “nonlinearity is useless”; within the framework the residual branch still outperforms the linear base on the S&P 500 at $T = 24$ across both seeds, and those two statements are consistent. **Sample size.** The asset-level association rests on nine assets and one seed per asset (two for GSPC); confidence intervals for association coefficients are wide and the analysis is exploratory. **Regime definition.** Regimes are volatility quantiles of a causal window measure; the calm-regime strata are statistically indistinguishable between operators, so “calm” is an acceptance result rather than a separate effect. **Environment.** Runs are CPU-only; wall-clock times differ from published GPU numbers and only relative comparisons are interpreted.

10. Conclusion

The reproduction reproduces the published ordering at $T = 24$ and reproduces it robustly across seeds. The diagnostics that follow refine, but do not overturn, that result: the marginal value of the nonlinear branch is asset-dependent rather than regime-universal, the volatility gate does not pay, and matched-capacity comparisons show that the earlier uniform preference for the linear operator was partly a capacity artefact. No robust experimental basis for sample-level routing was found, whereas genuine asset-level operator heterogeneity was. The next question the evidence supports is therefore not “how should a router choose?” but “in which reproducible contexts is a stronger nonlinear operator worth its capacity?”, to be answered against a capacity-matched baseline.

Appendix A. Reproducibility notes

All reproduction assets live under `reproduction/` in the project repository: analysis scripts in `reproduction/analysis/`, batch runners in `reproduction/batches/`, and per-setting results in `reproduction/results/`. Representative commands:

```
# single run (CPU)
.venv\Scripts\python.exe run.py --task_name long_term_forecast --is_training 1 \
  --model_id GSPC_96_24 --model DeReFusion --data custom --root_path ./dataset/ \
  --data_path GSPC-2016-2025.csv --features MS --target Close --freq b \
  --seq_len 96 --label_len 48 --pred_len 24 --enc_in 4 --dec_in 4 --c_out 1 \
  --d_model 32 --moving_avg 25 --train_epochs 30 --batch_size 32 \
  --learning_rate 0.0001 --patience 5 --lradj cosine --rand_seed 2021 --no_use_gpu

# regime stratification (causal, primary measure = relative volatility)
.venv\Scripts\python.exe reproduction/analysis/analyze_volatility_regimes.py \
  --csv dataset/GSPC-2016-2025.csv --tag GSPC --seed 2021 --rv-mode relative
```

Four environment facts cost real time and are recorded for reuse: the data pipeline imports `patool`, `huggingface_hub`, `sktime` and `datasets` unconditionally; `joblib` must be pinned to 1.5.3; CPU runs require `-no_use_gpu`; and killing `DataLoader` worker processes (`spawn_main`) deadlocks the parent trainer, so a stalled run must be detected by process CPU time rather than by log output.

References

- [1] Hsieh, C.-Y., Chen, Y.-C. (2026). DeReFusion: a decomposition-reconstruction fusion framework for financial time series forecasting. *Applied Soft Computing* 203, 116252.
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